Fairway Accounting Group

Cloud and Security Architecture

1. Purpose

This document provides two additional architecture diagrams for the Fairway Accounting Group Security Overhaul project.

The first diagram presents a proposed cloud security design that Fairway could adopt in the future. The second diagram shows the main security controls implemented and tested in the capstone lab.

These diagrams supplement the original and secured network diagrams already included in the main security audit report.

2. Proposed Cloud Architecture

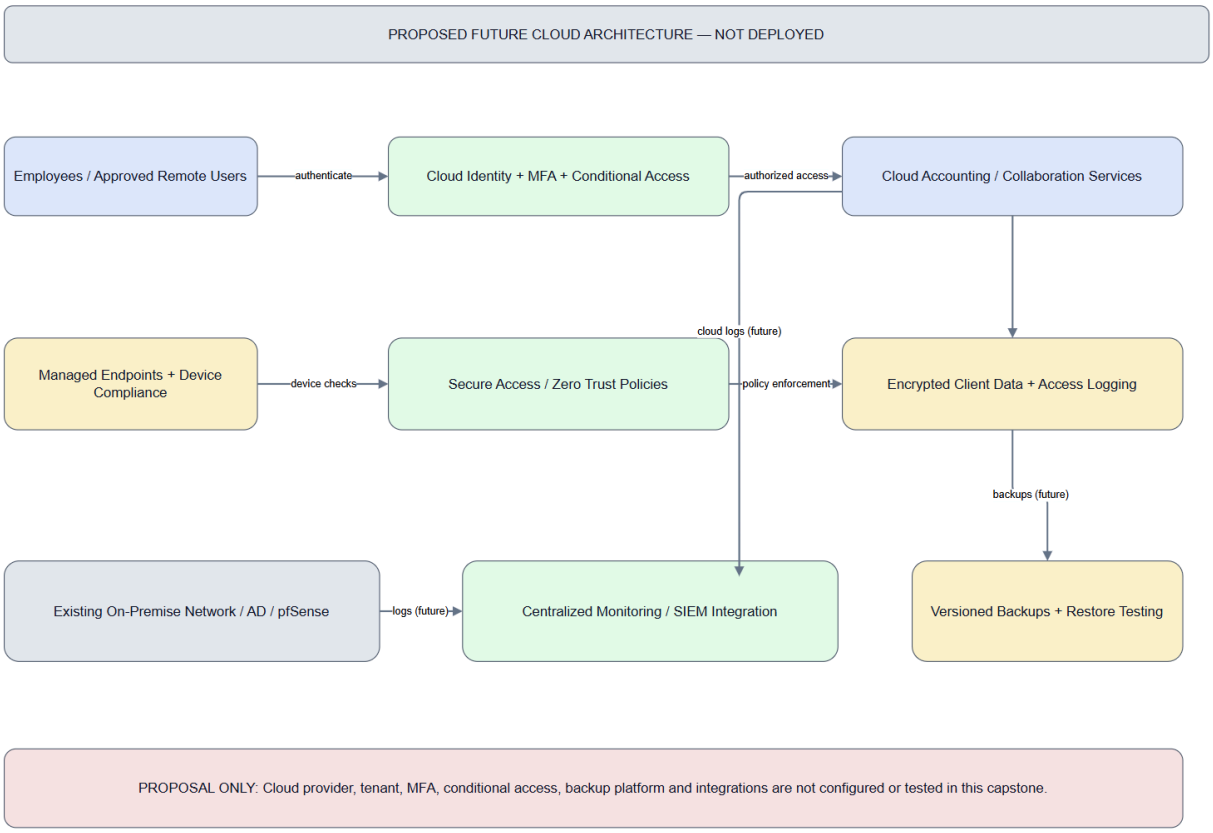


Figure 1. Proposed cloud architecture for Fairway Accounting Group. This design represents future recommendations and was not deployed during the project.

As Fairway continues to grow, cloud services could help employees securely access accounting applications, collaborate on documents, and protect client information.

The proposed architecture includes multifactor authentication, device compliance checks, access policies, encrypted data storage, centralized logging, and backup protection.

Before implementing these services, Fairway would need to select a cloud provider, review its compliance requirements, configure the security controls, and test access and recovery procedures.

3. Implemented Security Architecture

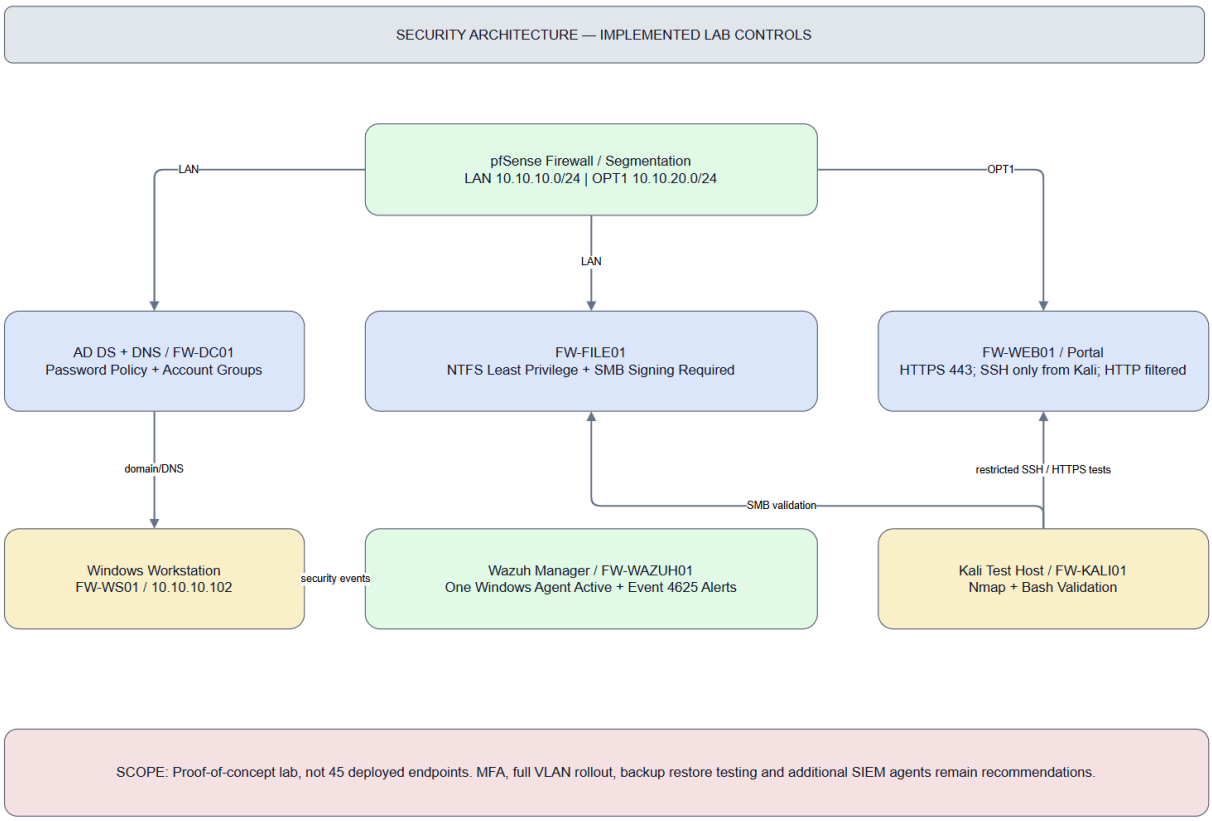


Figure 2. Security architecture showing the primary controls configured and tested in the virtual lab.

The lab uses pfSense to connect and control traffic between the internal network and a separate web server segment.

Windows Server provides Active Directory, DNS, and file services. Security improvements include stronger domain password requirements, corrected group membership, restricted file access, and required SMB signing.

The client portal uses HTTPS, with SSH access restricted to the Kali testing machine. Wazuh provides centralized monitoring for one Windows workstation, including detection of failed authentication attempts.

Nmap and a Bash validation script were used to test selected network services and document results.

4. Implemented Controls and Future Recommendations

Implemented and Tested	Recommended for Future Deployment
pfSense firewall and web server segmentation	Additional VLANs and internal segmentation
Active Directory access controls	Multifactor authentication
Domain password policy improvements	Cloud identity and conditional access
HTTPS client portal	Production-trusted TLS certificates
Wazuh monitoring of one workstation	Monitoring across all important endpoints
Nmap and Bash validation	Scheduled vulnerability assessments
SMB signing requirements	Tested backup and recovery procedures

5. Conclusion

These diagrams show both the security improvements completed in the Fairway lab and a possible direction for future development.

The implemented architecture demonstrates practical configuration and testing experience. The proposed cloud architecture provides a starting point for expanding security as the company grows.

The proposed controls would require additional planning, implementation, and validation before they could be considered operational.